One of the crucial parts was the contour nozzle. After convincing a committee to use this French-made nozzle without subjecting it to a high-altitude test, we had managed to get three nozzles from Volvo, Sweden, thanks to old friends in France.

Most of the nozzles have a convergent part, a divergent part and a neck that attaches the two. The gas generated in the combustion chamber gets expanded in the nozzle to convert pressure into kinetic energy that would lift the rocket. Nozzles are designed keeping in mind the pressure generated by the combustion and the atmospheric pressure outside.

Here was a small hitch. The atmospheric pressure drops as the altitude goes up. This means a nozzle that works fine at sea level may not optimally utilize the pressure generated at a high altitude where the atmospheric pressure is low. A PSLV has four stages and hence needs four nozzles for the four stages; each one is designed to work optimally for its entire flight duration. Often, fixed diameter nozzles result in waste of pressure and underperformance.

The Vikas engine nozzle is a versatile piece of engineering. Unlike the nozzle of the first-stage engine, which has a conical divergent part, this one is bell-shaped. While the other nozzles are more or less of the same thickness, the bell-shaped nozzle is also of tapering thickness, which means it is thickest (about 5mm) at the neck end and progressively becomes thin to be the thinnest (about 1.5mm) at the divergent end. It is designed to bring down the weight without compromising on the performance at high altitude where the pressure dips drastically.

The special design entails special processes. While a normal, conical nozzle is easy to make, using the roll-and-weld method, the Vikas engine nozzle is made using what we call the ‘flow technique’. In simple terms, it means the nozzle is made by making the alloy flow, while high-precision machines ‘pull and mould’ it into the bell-jar shape with tapering thickness. To make it simpler, think about a potter’s wheel, only replace the clay with alloy and the potter’s hands with the high-precision machines.

To get the nozzle right, the machines, method and the material—in this case a cobalt alloy that the French called KC-20WN, and the Americans